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## Simultaneously prediction of multiple wheat leaf phenotypes using hyperspectral imaging with multi-task machine and deep learning

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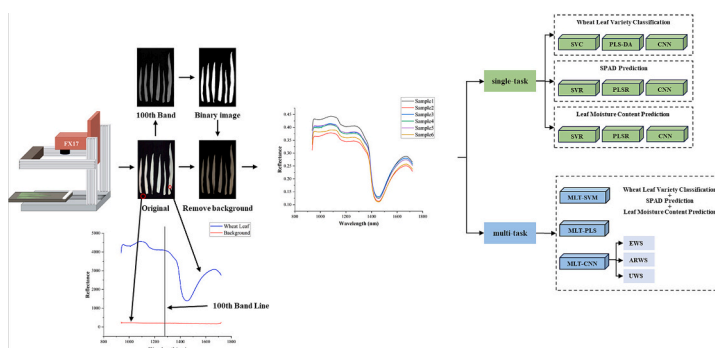
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### HIGHLIGHTS

- Wheat variety, water content and SPAD were predicted by hyperspectral imaging.
- Single-task models performed well on regression and classification of phenotypes.
- Multi-task models performed well on classification and regression of phenotypes.
- Different loss function of multi-task deep learning models showed promising results.

### GRAPHICAL ABSTRACT



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### ABSTRACT

Chlorophyll content (represented by the Soil and Plant Analyzer Development (SPAD) value) and leaf moisture content (LMC) are two key physiological phenotypic traits during wheat growth, and their variations among different wheat varieties reflect crop growth, stress response, and breeding evaluation. Simultaneous identification of wheat varieties and prediction of SPAD and LMC are therefore important for precision crop monitoring. In this study, hyperspectral imaging was employed to acquire leaf spectral information from four wheat varieties. After spectral preprocessing and outlier screening, 684 valid samples were retained for model development and evaluation. Single-task and multi-task models were constructed for wheat variety classification, SPAD prediction, and LMC prediction using support vector machine (SVM), partial least squares (PLS), convolutional neural network (CNN), and multi-task CNN (MLT-CNN) algorithms. In the MLT-CNN, a shared one-dimensional spectral feature extraction module and three task-specific branches were designed, and equal weight strategy (EWS), adjustable regularization weighted strategy (ARWS), and uncertainty-based weighted strategy (UWS) were compared. The best single-task models achieved a test-set classification accuracy of 0.75, with correlation coefficients ( $r$ ) of 0.83 for SPAD and 0.84 for LMC. The MLT-CNN with EWS achieved balanced test-set performance across the three tasks, with a classification accuracy of 0.69, an  $r$  value of 0.84 for SPAD, and an  $r$  value of 0.81 for LMC. Shapley additive explanations (SHAP)-based visualization was further performed for both single-task

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CNNs and MLT-CNN task branches to identify important wavelengths and interpret shared and task-specific spectral contributions. These results indicate that hyperspectral imaging combined with multi-task learning provides a feasible and interpretable spectroscopic strategy for integrated wheat leaf phenotyping.

## 1. Introduction

Wheat is one of the most important staple crops worldwide and plays a central role in global food security. It is widely cultivated and consumed across different regions and serves as an essential source of dietary energy and plant-based protein. In China, wheat, rice, and maize are considered the three major food crops [1].

In crop growth monitoring and management, leaves are important carriers of plant health information. The morphological traits of wheat leaves can intuitively reflect growth conditions, whereas physiological traits provide information on internal metabolic activities. Among these traits, chlorophyll-related SPAD values and leaf moisture content (LMC) are closely associated with crop nutritional status, photosynthetic activity, water balance, and stress response. Variations in chlorophyll-related traits are sensitive indicators of nutrient levels and photosynthetic capacity across different growth stages, and they are widely used to evaluate nitrogen status and photosynthetic potential [2]. The Soil and Plant Analyzer Development (SPAD) value is positively correlated with chlorophyll content and is therefore commonly used as a rapid, non-destructive proxy for chlorophyll-related variation in leaves [3]. LMC is closely related to water balance, drought resistance, and physiological activity, and is regarded as an important parameter for evaluating crop stress tolerance and vitality. In precision agriculture, SPAD and LMC are widely used for diagnosing crop nutritional status, monitoring stress responses, and supporting yield prediction. Different wheat varieties may exhibit cultivar-dependent physiological and biochemical characteristics, including differences in pigment accumulation, water regulation, photosynthetic efficiency, and spectral reflectance responses. Therefore, modeling SPAD and LMC from a cultivar-specific perspective can provide insights into phenotypic differences among wheat varieties and support more refined wheat growth monitoring and breeding evaluation.

Traditional detection methods for leaf physiological traits often rely on destructive sampling and laboratory chemical analysis, which are time-consuming and unsuitable for rapid large-scale monitoring. Hyperspectral imaging provides a non-destructive alternative by integrating imaging technology with spectral analysis, enabling the simultaneous acquisition of spatial and spectral information from plant samples. In the visible, near-infrared, and short-wave infrared (SWIR) regions, leaf spectral responses are closely related to pigment absorption, water absorption, tissue structure, and biochemical constituents. In particular, SWIR spectra are influenced by overlapping absorption and scattering processes associated with water status, internal leaf structure, organic compounds, and O—H, C—H, and N—H bond vibrations. Therefore, wavelength-level interpretation is necessary to clarify how different spectral regions contribute to cultivar classification and physiological trait prediction. By continuously acquiring and stacking spectral images across multiple wavelengths, hyperspectral imaging can generate a three-dimensional data cube containing two-dimensional spatial information and one-dimensional spectral information [4]. In recent years, this technique has been widely applied in plant phenotyping, including salt stress prediction [5], chlorophyll-related trait estimation, and species classification [6]. In wheat research, hyperspectral imaging has also been used for detecting nitrogen status and water-related traits [7].

Recent studies have demonstrated the broad applicability of hyperspectral imaging combined with machine learning and deep learning for crop monitoring and plant phenotyping. A recent review summarized the use of hyperspectral imaging and artificial intelligence algorithms in crop disease detection, crop classification, nutrient estimation, variety

identification, and plant phenotyping across different crops, spectral ranges, and modeling strategies [8]. In wheat research, the Hyper-Leaf2024 dataset provided hyperspectral images of wheat flag leaves with associated classification and regression targets, including cultivar, fertilizer content, yield, stomatal conductance, and chlorophyll fluorescence, highlighting the potential of leaf-level hyperspectral imaging for both categorical and continuous phenotypic tasks [9]. In addition, UAV-based hyperspectral imagery has been used to estimate leaf area index and chlorophyll traits in wheat for yield prediction, further demonstrating the value of hyperspectral data for wheat physiological trait assessment [10]. Beyond wheat, hyperspectral or multispectral imaging has also been applied to maize chlorophyll estimation, maize nitrogen deficiency detection and crop trait quantification, and the evaluation of chemical traits and water status in avocado, olive, and grape leaves, indicating that spectral imaging has been widely explored across different crops and study regions [11–13]. However, most existing studies have focused on single-target classification, single-trait regression, or the joint prediction of multiple continuous traits. For example, deep multi-task regression has been used to simultaneously predict SPAD and water content in rice leaves, but cultivar classification was not integrated into the same modeling framework [14]. Recent studies have shown that explainable artificial intelligence methods such as Shapley additive explanations (SHAP) can be used to identify informative spectral bands and explain the contribution of wavelengths in hyperspectral image classification [15]. Moreover, many existing hyperspectral studies mainly emphasize prediction accuracy, whereas the wavelength-level contribution patterns underlying different phenotyping tasks are often insufficiently interpreted. In particular, it remains necessary to clarify which wavelength regions contribute to cultivar classification, SPAD prediction, and LMC prediction, and whether these tasks rely on shared spectral information or task-specific wavelength responses. Similar SHAP-based interpretations have recently been used to connect hyperspectral wavelength contributions with plant physiological responses, supporting the value of explainable spectral modeling [16]. Therefore, studies that simultaneously integrate cultivar classification, SPAD prediction, LMC prediction, and wavelength-level spectral interpretation within a unified leaf-level hyperspectral multi-task framework remain comparatively limited. This gap motivates the development of an integrated and interpretable multi-task spectral modeling strategy for wheat leaf phenotyping.

Conventional approaches usually focus on single-task modeling, in which classification or regression models are developed and optimized separately. Although such models can achieve reliable performance for individual targets, they do not fully exploit the potential relationships among different phenotypic traits. Traditional machine learning methods, such as support vector machine (SVM) and Partial Least Squares (PLS), often achieve reliable performance through statistical modeling and feature-space projection, especially for small-to-medium-sized spectral datasets. Deep learning methods, by contrast, can automatically learn nonlinear spectral representations and have shown strong potential for high-dimensional hyperspectral modeling. However, their performance depends on data scale, model architecture, task complexity, and training strategy. Multi-task learning has recently attracted increasing attention because it enables multiple related tasks to be modeled within a unified framework and allows shared representations to be learned across tasks [17]. In wheat leaf phenotyping, cultivar identity, SPAD, and LMC are physiologically related but correspond to different modeling objectives, including classification and regression. A single leaf spectrum represents a mixed response of cultivar-dependent structure, pigment-related physiological status,

water status, and biochemical composition. Therefore, integrating wheat variety classification, SPAD prediction, and LMC prediction within one framework may reduce the need to construct separate models and support a more comprehensive interpretation of cultivar-related physiological variation. This integrated multi-task spectral modeling strategy constitutes one of the major contributions of this study.

Within multi-task learning frameworks, loss-function design is critical because classification and regression tasks differ substantially in optimization objectives, gradient propagation, loss scales, and convergence behavior. To investigate how task weighting affects multi-task spectral modeling, three loss-weighting strategies were compared in this study: equal weight strategy (EWS), adjustable regularization weighted strategy (ARWS), and uncertainty-based weighted strategy (UWS). These strategies were used to evaluate different ways of coordinating task contributions during training and to identify a suitable balance among wheat variety classification, SPAD prediction, and LMC prediction. In addition to predictive performance, model interpretability is also important for clarifying whether the multi-task model uses shared spectral regions, task-specific spectral cues, or both. Therefore, SHAP-based wavelength contribution analysis was introduced in this study to quantify the contribution of each input wavelength to the CNN-based model outputs and to provide a spectroscopic explanation for the shared feature extraction module and task-specific branches.

Four wheat varieties (Nonglinmai No.1, Zhehangmai No.9, Jiamai No.2211, and Zhexiangmai No.1) were selected in this study. Hyperspectral imaging was combined with machine learning and deep learning models to identify wheat cultivar types and predict SPAD and LMC from leaf spectra. The specific objectives were:

(1) to construct single-task models, including Support Vector Classification (SVC), Partial Least Squares Discriminant Analysis (PLS-DA), and Convolutional Neural Network (CNN) for wheat variety classification, as well as Support Vector Regression (SVR), Partial Least Squares Regression (PLSR), and CNN for SPAD and LMC prediction;

(2) to establish and compare multi-task models, including multi-task SVM (MLT-SVM), multi-task PLS (MLT-PLS), and multi-task CNN (MLT-CNN), for simultaneous wheat variety classification, SPAD prediction, and LMC prediction;

(3) to compare different loss-weighting strategies in the MLT-CNN framework and evaluate their influence on task balance and overall multi-task performance.

(4) to conduct SHAP-based wavelength contribution analysis to identify task-related spectral regions and interpret the shared and task-specific wavelength information used by the CNN-based models.

## 2. Materials and methods

### 2.1. Sample collection

Fresh wheat leaf samples were collected from the experimental field of Huzhou Academy of Agricultural Sciences. Four wheat varieties were selected including Nonglinmai No.1, Zhehangmai No.9, Jiamai No.2211, and Zhexiangmai No.1. To improve model robustness and cover a wider range of physiological variation, leaves with different greenness levels, including dark green, yellow-green, and yellowish leaves, were selected from each cultivar. Sampling was conducted at 7:00 a.m. on two consecutive days to reduce the influence of diurnal variation in photosynthetic activity on chlorophyll-related traits. 90 fresh leaves were collected from each cultivar per day, resulting in 180 leaves per cultivar and 720 samples in total.

### 2.2. Hyperspectral data acquisition

The hyperspectral imaging system consisted of a light source, a hyperspectral camera, an electronically controlled mobile platform, and a desktop computer. A Specim FX17 camera (Spectral Imaging Ltd., Oulu, Finland) was used to acquire hyperspectral data in the SWIR

region of 900–1700 nm. Data acquisition was controlled using Lumo software. Before image acquisition, the height of the lens and light source was adjusted to obtain clear leaf images and uniform illumination. The distance between the camera lens and the sample surface was approximately 310 mm. The sample plate had dimensions of 400 mm × 200 mm (length × width), and the camera support system had dimensions of 500 mm × 325 mm × 340 mm (length × width × height). The mobile platform speed was set to 40 mm/s, and the scanning speed was set to 24.7 mm/s to reduce image distortion.

During acquisition, wheat leaves were placed horizontally on the mobile platform, with six leaves scanned at a time. The leaves were numbered sequentially from bottom to top (1–6) to maintain correspondence between hyperspectral images and physiological measurements. All images were acquired in a darkroom to minimize external light interference and improve the stability of spectral acquisition.

After hyperspectral image acquisition, the raw hyperspectral images were corrected using black and white reference calibration according to the following equation:

$$R = \frac{R_0 - B}{W - B} \quad (1)$$

where  $R_0$  represents the raw image,  $B$  is the black reference (obtained by covering the lens),  $W$  is the white reference (measured with a standard white board), and  $R$  is the corrected spectral reflectance.

### 2.3. Spectral data extraction and preprocessing

The reflectance image at each band was examined, and the image at the 100th band was found to provide clear discrimination between wheat leaves and the background. Therefore, the grayscale image at the 100th band of the raw hyperspectral data was extracted for mask generation. For image segmentation, threshold binarization was applied by setting pixels below the threshold to 0 and pixels above the threshold to 1, thereby generating a binary image containing the leaf regions and background, as shown in Fig. 1. Subsequently, connected-component analysis was performed to label the leaf regions in the binary image, preserve the effective leaf area, and extract the centroid of each leaf region. The identified leaf regions were then sorted from right to left according to their centroid coordinates along the X-axis to maintain consistency with the sample numbering order.

For each identified leaf region, a binary mask with the same spatial dimensions as the hyperspectral image was generated. The mask was then multiplied element-wise with the hyperspectral data to isolate the spectral information of each leaf. For each spectral band, the sum of valid pixel values within the masked region was divided by the number of valid pixels to obtain the mean reflectance. The mean reflectance values across all bands were saved as CSV files and used as the spectral dataset for subsequent modeling.

Hyperspectral images are often affected by instrumental noise, environmental variation, band redundancy, and mixed noise, which may reduce data quality and model stability. Recent studies have systematically reviewed hyperspectral image denoising methods from model-driven, data-driven, and model-data-driven perspectives, and have further developed advanced unsupervised and spatial-spectral deep learning methods for HSI denoising [18–20]. These studies indicate the importance of noise suppression in hyperspectral image processing. In this study, since the objective was to extract one-dimensional mean spectra from segmented wheat leaf regions rather than to reconstruct full hyperspectral images, a lightweight spectral preprocessing strategy was adopted. Specifically, the first 10 bands and the last 14 bands were removed to reduce the influence of low signal-to-noise bands at the spectral boundaries. The remaining 200 consecutive bands from band 11 to band 210 were retained as input features. A five-point moving average filter was then applied to smooth the spectra, reduce random fluctuations, and preserve the main spectral characteristics.

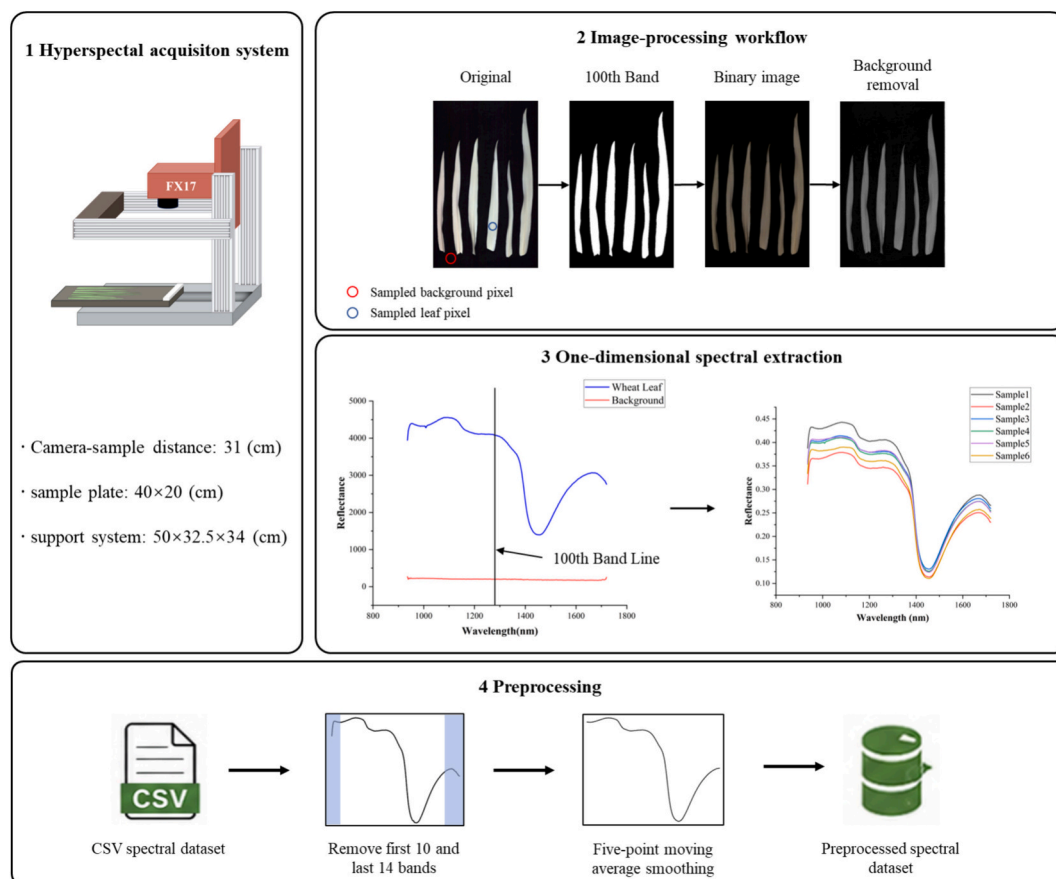


Fig. 1. Spectral data extraction flowchart.

#### 2.4. Measurement of SPAD and LMC

The SPAD value, used as a chlorophyll-related indicator, was measured using a SPAD chlorophyll meter (Konica Minolta, Osaka, Japan). The instrument was calibrated prior to use. Each leaf was kept in the dark immediately after collection and measured directly after hyperspectral imaging to ensure consistency between spectral acquisition and physiological measurement. For each leaf, five points were randomly measured following the order of hyperspectral acquisition, and the mean value was used as the final SPAD value of that leaf.

After SPAD measurement, the fresh weight (FW) of each leaf was measured using a precision electronic balance (Sartorius BSA223S, Göttingen, Germany). The leaves were then placed into labeled envelopes and oven-dried at 75 °C for two consecutive days, with 8 h of drying per day. The dry weight (DW) was subsequently measured, and LMC was calculated as:

$$LMC = \frac{FW - DW}{FW} \times 100 \quad (2)$$

#### 2.5. Algorithm

To perform classification and regression modeling for hyperspectral wheat leaf data, two types of models were constructed in this study: single-task models and multi-task models. The single-task models were developed to perform either wheat variety classification or physiological trait prediction independently, whereas the multi-task models were designed to simultaneously perform wheat variety classification, SPAD prediction, and LMC prediction.

For model development, the dataset was divided into calibration, validation, and test sets. The calibration set was used for model training and hyperparameter tuning. For SVM and PLS models, candidate

hyperparameters were selected using cross-validation within the calibration set. The validation set was used for model selection and performance monitoring, whereas the test set was kept independent and used only for final model evaluation. This strategy was adopted to avoid using test-set information during hyperparameter optimization or model selection.

##### 2.5.1. Single-task model

In this study, three classical modeling approaches, including SVM, PLS, and CNN, were employed for wheat variety classification, SPAD prediction, and LMC prediction from hyperspectral wheat leaf data.

SVM is a supervised learning algorithm based on the construction of an optimal separating margin and has been widely applied to classification and regression tasks [21]. Its core idea is to improve model generalization by constructing a hyperplane that maximizes the inter-class margin. Meanwhile, kernel functions can be used to map data into a high-dimensional feature space to handle nonlinear relationships. In this study, SVC was used to build the wheat variety classification model, while SVR was used to build regression models for SPAD and LMC. The radial basis function (RBF) kernel was used. The penalty coefficient  $C$  and kernel coefficient  $\gamma$  were optimized by cross-validation within the calibration set, with the search range set to  $[10^{-7}, 10^7]$ .

PLS is a multivariate statistical modeling method that integrates dimensionality reduction and variable selection, and is particularly suitable for high-dimensional feature data with strong collinearity [22]. In this study, PLSR was used to build regression models for SPAD and LMC. For classification, PLS-DA was adopted. The categorical dependent variables were first transformed by dummy variable encoding, and the classification task was then solved as a regression problem. The class corresponding to the maximum predicted value was taken as the classification output. The optimal number of latent components was

determined by cross-validation within the calibration set. The selected parameter was then used to train the final model, and the validation set was used for model selection before final evaluation on the independent test set.

CNN is a deep learning model with strong feature extraction capability and is suitable for automatic nonlinear modeling of structured inputs [23]. In this study, one-dimensional CNN models were constructed using spectral sequences as input features. For the classification task, the cross-entropy (CE) loss function was used. The model contained one convolutional layer with a kernel size of 5, a batch size of 64, and a learning rate (LR) of 0.0001. The convolutional layer served as a feature extractor. After feature extraction, the multi-channel features were flattened into a one-dimensional vector and then fed into fully connected layers for feature integration and classification. The final output layer mapped the features to four wheat variety categories. The structure of the classification model is shown in Fig. 2(a).

For the regression tasks, the mean squared error loss function was used. The regression CNN contained four convolutional layers with a kernel size of 3 and a stride of 3, a batch size of 32, and an LR of 0.001. The structure and function of the regression model were similar to those of the classification model, except that the output layer was used for continuous-value prediction. The structure of the regression model is shown in Fig. 2(b). During training, the Adam optimizer was employed to improve training stability and accelerate convergence.

### 2.5.2. Multi-task model

To further achieve joint modeling of the three tasks, namely wheat variety classification, SPAD prediction, and LMC prediction, three multi-task learning models were constructed in this study: MLT-SVM, MLT-PLS, and MLT-CNN. These models were developed to improve modeling efficiency and explore the potential relationships among cultivar identity, SPAD, and LMC from leaf spectral features.

MLT-SVM consisted of one SVC classifier and two SVR regressors within a unified input feature space. In the parameter optimization process, the candidate kernel functions were set as ['linear', 'rbf', 'poly']; the search range of the penalty coefficient C was [ $10^{-7}$ ,  $10^7$ ]; and the kernel coefficient gamma was selected from ['scale', 'auto']. To comprehensively evaluate the overall performance of the three tasks, a joint scoring metric was introduced as follows:

$$\text{HybridScore} = \text{accuracy} + r_1 + r_2, \quad (3)$$

where accuracy represents the classification accuracy, and  $r_1$  and  $r_2$  represent the Pearson correlation coefficients ( $r$ ) for SPAD and LMC prediction, respectively. During parameter optimization, candidate parameter combinations were evaluated using this joint score, and the parameter combination with the highest score was selected for the final model.

MLT-PLS was designed to simultaneously perform one classification task and two regression tasks by sharing the same input feature space and latent-variable representation. Specifically, one PLS-DA classifier and two PLSR regressors were constructed for wheat variety classification, SPAD prediction, and LMC prediction, respectively. Similar to MLT-SVM, MLT-PLS used the joint scoring metric for parameter selection. This design preserved the dimensionality-reduction advantage of PLS while allowing the three tasks to be evaluated within a unified modeling framework.

MLT-CNN served as the core multi-task deep learning model in this study. Its structure consists of a shared feature extraction module and three task-specific branches. The shared feature extraction module adopted a one-dimensional convolutional structure to extract representative spectral features from the input spectral sequence. Two convolutional layers were designed, with a kernel size of 7 and strides of 3 and 2, respectively. The three task-specific branches corresponded to one classification output for wheat variety identification and two regression outputs for SPAD and LMC prediction. Each branch contained

a fully connected layer. The model was trained in an end-to-end manner using the Adam optimizer, with a LR of 0.001 and a batch size of 8. LR decay was applied with a patience of 50 and a factor of 0.8. The model structure is shown in Fig. 2(c). In the two regression branches, Regressor 1 was used for SPAD prediction, and Regressor 2 was used for LMC prediction. To investigate the influence of task weighting on multi-task learning performance, three loss-weighting strategies were designed and compared as follows:

1. EWS: The losses of the three tasks were assigned equal weights, and the total loss was calculated as the weighted sum:

$$\text{loss} = \frac{1}{3} \text{loss}_{\text{reg1}} + \frac{1}{3} \text{loss}_{\text{reg2}} + \frac{1}{3} \text{loss}_{\text{cls}} \quad (4)$$

2. ARWS: Three task weights ( $\alpha$ ,  $\beta$ ,  $\gamma$ ) were introduced as learnable parameters. The loss function was defined as:

$$\text{loss} = \alpha \text{loss}_{\text{reg1}} + \beta \text{loss}_{\text{reg2}} + \gamma \text{loss}_{\text{cls}} + \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} \quad (5)$$

This strategy adaptively adjusts task weights while introducing a regularization term to prevent extreme weight allocation and alleviate task imbalance. The values of  $\alpha$  and  $\beta$  vary within (0,1), and  $\gamma$  is calculated using the following equation to ensure that the sum of the three task weights equals 1:

$$\gamma = 1 - \alpha - \beta \quad (6)$$

3. UWS: Task uncertainty was incorporated into the loss function for joint optimization. The total loss was defined as:

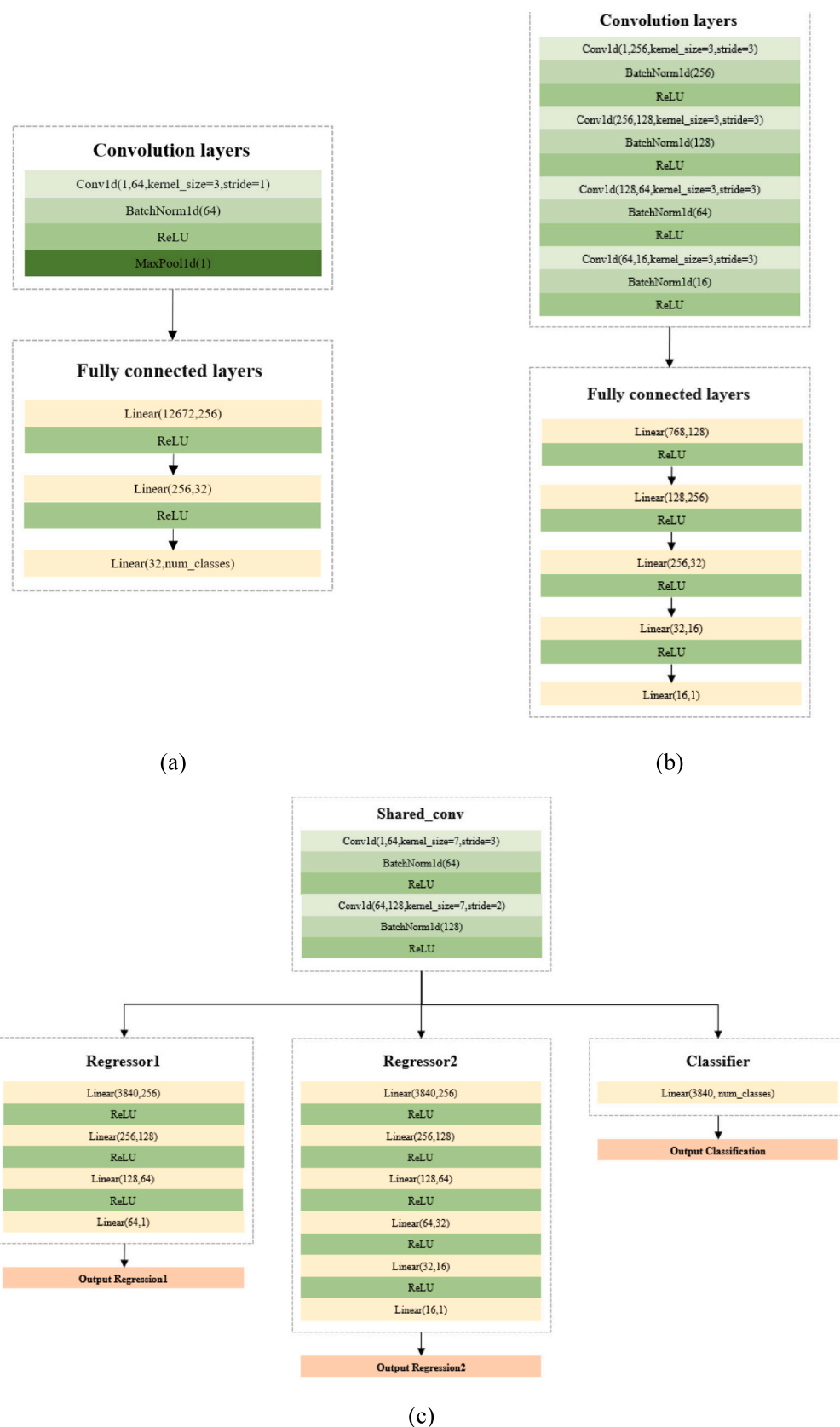
$$\begin{aligned} \text{loss}_{\text{total}} = & \frac{1}{2\sigma_{\text{cls}}^2} L_{\text{cls}}(W) + \log \sigma_{\text{cls}} + \frac{1}{2\sigma_{\text{reg1}}^2} L_{\text{reg1}}(W) + \log \sigma_{\text{reg1}} \\ & + \frac{1}{2\sigma_{\text{reg2}}^2} L_{\text{reg2}}(W) + \log \sigma_{\text{reg2}} \end{aligned} \quad (7)$$

where the weighting coefficients for each task are determined by the uncertainty parameters  $\sigma_i$ , which are learned during model training. In the loss function, the terms  $\frac{1}{2\sigma_i^2}$  and  $\frac{1}{\sigma_i^2}$  are used to scale the corresponding task losses, with larger task uncertainty leading to a smaller loss weight. In addition,  $\log \sigma_i$  serves as a regularization term to prevent  $\sigma_i$  from becoming excessively small and causing training instability. This strategy enables adaptive adjustment of the relative influence of different tasks when they exhibit different noise levels and learning difficulties, thereby contributing to training stability [24].

### 2.6. SHAP-based wavelength contribution analysis

To further interpret the wavelength-level contribution of the CNN-based models, SHAP analysis was performed for both the single-task CNNs and the MLT-CNN. For the single-task CNNs, SHAP values were calculated separately for wheat variety classification, SPAD prediction, and LMC prediction. For the MLT-CNN, the trained multi-task model was wrapped into three task-specific output models, corresponding to the classification branch, SPAD branch, and LMC branch, and SHAP values were calculated separately for each branch.

For classification outputs, the logit corresponding to the predicted class was used as the target output for SHAP calculation. For regression outputs, the continuous predicted value was used directly as the target output. A subset of training samples was used as the background reference set, and the test samples were used as the explanation set. For each task, the absolute SHAP values were averaged across the test samples to obtain the mean absolute SHAP value of each wavelength, which is commonly used to summarize global feature contribution patterns in SHAP analysis [25]. The resulting wavelength contribution curves were then normalized by the maximum value of each task to facilitate comparison among different outputs. The SHAP curves were used to analyze task-related wavelength contributions and to compare shared and task-specific spectral response patterns among wheat variety classification,



**Fig. 2.** Diagrams of the single-task and multi-task deep learning models. (a) Single-task CNN classification model; (b) single-task CNN regression model; (c) MLT-CNN model.

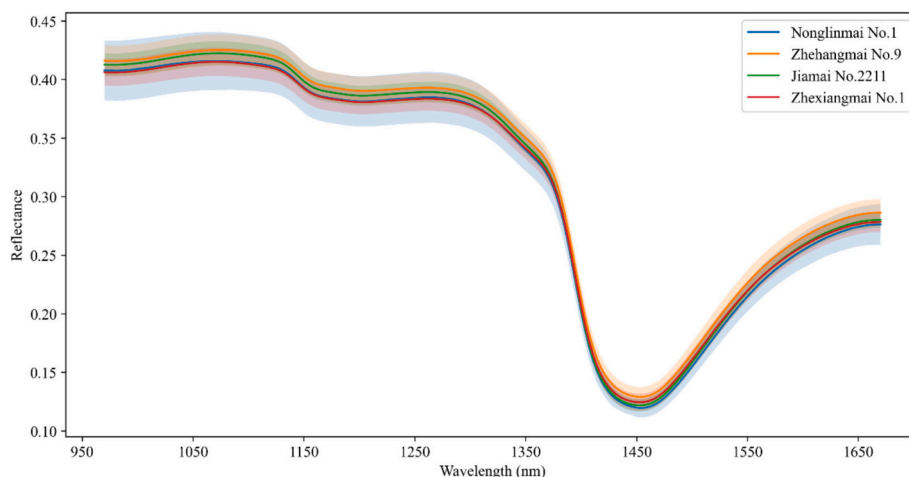


Fig. 3. Averaged near-infrared spectra and standard deviations of leaves from four wheat varieties.

SPAD prediction, and LMC prediction.

## 2.7. Evaluation metrics and computational environment

For model performance evaluation, accuracy was used as the evaluation metric for wheat variety classification. Accuracy was defined as the ratio of correctly predicted samples to the total number of samples and was used to evaluate classification performance. For regression tasks, the  $r$  and root mean square error (RMSE) were used to evaluate the prediction performance of SPAD and LMC. The  $r$  value measures the linear correlation between predicted and measured values, with a range of  $[-1, 1]$  and a value closer to 1 indicates better predictive performance. RMSE measures the average prediction error, with a lower value indicating better prediction accuracy.

All experiments were conducted on a Windows 11 64-bit operating system. The hardware configuration included an Intel Core i7-14700KF CPU and an NVIDIA GeForce RTX 4070 Ti SUPER GPU with 16 GB of memory. The program development and execution were carried out on PyCharm (version 2024.1) and Python (version 3.12.9) platforms. Models such as SVC, SVR, PLS-DA, PLSR, MLT-SVM, and MLT-PLS were built on Scikit-learn (version 1.6.0), while CNN and MLT-CNN models were constructed on PyTorch (version 2.5.1 + cu124).

## 3. Results

### 3.1. Spectra of wheat leaves from different varieties

The spectral characteristics of plant leaves are influenced by pigments, water status, tissue structure, and biochemical constituents. Therefore, leaf spectral reflectance can provide useful information on plant growth status and physiological variation. Fig. 3 shows the mean reflectance spectra of the four wheat varieties in the near-infrared/short-wave infrared range of 970–1670 nm, where the x-axis represents wavelength and the y-axis represents spectral reflectance. Overall, the spectra of the four varieties exhibited similar trends, indicating broadly similar leaf biochemical compositions and structural characteristics. Nevertheless, differences in reflectance magnitude and absorption features were observed among varieties, suggesting cultivar-dependent variation in leaf physiological and biochemical properties. The characteristic absorption valleys and reflectance differences are mainly associated with the first and second overtones, combination bands, and bending vibrations of C—H, O—H, and N—H bonds, which are related to water, protein, starch, and other biochemical constituents in wheat leaves [26,27].

### 3.2. Analysis of SPAD and LMC

Before final model development, outlier samples that might compromise data reliability were removed as a preliminary data-quality control step. These abnormal samples may have been caused by measurement errors, sample damage, abnormal physiological status, or mismatch between hyperspectral acquisition and physiological measurements. First, based on the original dataset, PLSR was used to visually examine the spectral-response relationships for SPAD and LMC, respectively. Then, outliers were identified by analyzing the differences between the PLSR-predicted values and the measured values. Subsequently, 18 SPAD-related outliers and 18 LMC-related outliers were removed, resulting in 36 outliers in total. This procedure was performed before final model development and was not used for hyperparameter optimization or final model selection. Ultimately, 684 valid samples were retained for subsequent modeling.

The retained dataset was divided into a calibration set (456 samples), a validation set (114 samples), and a test set (114 samples) at a ratio of 3:1:1. Care was taken to ensure that there were no duplicate samples across the three sets and that the sample distribution was relatively uniform within each set. For SPAD, the calibration set ranged from 32.75 to 57.80 (mean = 46.11), the validation set ranged from 34.42 to 56.55 (mean = 45.97), and the test set ranged from 35.45 to 57.70 (mean = 46.85). For LMC, the calibration set ranged from 69.27 to 81.27 (mean = 75.81), the validation set ranged from 69.83 to 80.08 (mean = 75.74), and the test set ranged from 69.26 to 81.97 (mean = 75.89). Sample statistics for each variety across different sets are detailed in Table 1.

The distribution of SPAD and LMC values showed that the retained samples covered a broad physiological range across the four wheat varieties. Specifically, Jiamai No.2211 exhibited relatively high SPAD values, whereas Zhehangmai No.9 showed relatively low SPAD values. For LMC, Nonglinmai No.1 exhibited relatively high values, whereas Zhexiangmai No.1 showed relatively low values. These differences indicate cultivar-dependent variation in the two key physiological phenotypic indicators and support the necessity of cultivar-specific and multi-task spectral modeling.

### 3.3. Performance of single-task and multi-task models

#### 3.3.1. Results of single-task models

In this study, single-task classification and regression models were developed to independently predict wheat variety, SPAD, and LMC from wheat leaf spectra. For model development and comparison, traditional machine learning models, including SVC, SVR, PLS-DA, and PLSR, and

**Table 1**  
SPAD and LMC of the collected wheat leaves.

| Variety          | Label | Dataset         | Sample number | Minimum | Maximum | Mean  | Standard deviation |
|------------------|-------|-----------------|---------------|---------|---------|-------|--------------------|
| Nonglinmai No.1  | SPAD  | Calibration Set | 113           | 33.55   | 57.45   | 47.01 | 5.44               |
|                  |       | Validation Set  | 28            | 36.32   | 54.93   | 45.10 | 5.16               |
|                  |       | Test Set        | 28            | 39.47   | 56.58   | 48.75 | 4.98               |
|                  | LMC   | Calibration Set | 113           | 72.77   | 81.27   | 77.02 | 1.60               |
|                  |       | Validation Set  | 28            | 72.96   | 79.20   | 76.36 | 1.59               |
|                  |       | Test Set        | 28            | 73.46   | 81.97   | 77.11 | 1.56               |
| Zhehangmai No.9  | SPAD  | Calibration Set | 113           | 32.75   | 56.25   | 44.57 | 4.64               |
|                  |       | Validation Set  | 28            | 34.42   | 53.53   | 44.52 | 4.46               |
|                  |       | Test Set        | 28            | 35.45   | 50.93   | 44.23 | 4.09               |
|                  | LMC   | Calibration Set | 113           | 70.53   | 81.16   | 75.75 | 1.77               |
|                  |       | Validation Set  | 28            | 72.05   | 77.66   | 75.45 | 1.54               |
|                  |       | Test Set        | 28            | 72.35   | 78.88   | 75.65 | 1.59               |
| Jiamai No.2211   | SPAD  | Calibration Set | 115           | 34.52   | 57.8    | 47.45 | 4.08               |
|                  |       | Validation Set  | 29            | 37.73   | 54.08   | 47.74 | 3.52               |
|                  |       | Test Set        | 29            | 38.42   | 57.70   | 49.03 | 4.96               |
|                  | LMC   | Calibration Set | 115           | 71.90   | 79.12   | 75.95 | 1.54               |
|                  |       | Validation Set  | 29            | 73.08   | 78.70   | 76.13 | 1.65               |
|                  |       | Test Set        | 29            | 72.53   | 80.18   | 76.39 | 1.87               |
| Zhexiangmai No.1 | SPAD  | Calibration Set | 115           | 32.98   | 55.02   | 45.40 | 4.19               |
|                  |       | Validation Set  | 29            | 37.12   | 56.55   | 46.44 | 4.33               |
|                  |       | Test Set        | 29            | 36.22   | 53.87   | 45.36 | 4.30               |
|                  | LMC   | Calibration Set | 115           | 69.27   | 78.87   | 74.55 | 1.94               |
|                  |       | Validation Set  | 29            | 69.83   | 80.08   | 75.02 | 1.96               |
|                  |       | Test Set        | 29            | 69.26   | 77.95   | 74.43 | 2.46               |

**Table 2**  
Results of single-task classification models.

| Model  | Dataset         | Accuracy    |
|--------|-----------------|-------------|
| SVC    | Calibration Set | <b>0.81</b> |
|        | Validation Set  | <b>0.77</b> |
|        | Test Set        | <b>0.75</b> |
| PLS-DA | Calibration Set | 0.74        |
|        | Validation Set  | 0.75        |
|        | Test Set        | 0.75        |
| CNN    | Calibration Set | 0.73        |
|        | Validation Set  | 0.74        |
|        | Test Set        | 0.68        |

**Table 3**  
Results of single-task regression models.

| Model | Dataset         | $r_1$       | RMSE | $r_2$       | RMSE |
|-------|-----------------|-------------|------|-------------|------|
| SVR   | Calibration Set | 0.84        | 2.63 | 0.84        | 1.04 |
|       | Validation Set  | 0.77        | 2.86 | 0.80        | 1.05 |
|       | Test Set        | 0.81        | 2.94 | 0.82        | 1.23 |
| PLSR  | Calibration Set | <b>0.86</b> | 2.43 | <b>0.87</b> | 0.96 |
|       | Validation Set  | <b>0.80</b> | 2.73 | <b>0.81</b> | 1.03 |
|       | Test Set        | <b>0.83</b> | 2.81 | <b>0.84</b> | 1.17 |
| CNN   | Calibration Set | 0.82        | 3.85 | 0.83        | 2.10 |
|       | Validation Set  | 0.74        | 3.99 | 0.77        | 2.04 |
|       | Test Set        | 0.80        | 4.39 | 0.84        | 2.11 |

Note:  $r_1$  is the  $r$  of SPAD,  $r_2$  is the  $r$  of LMC.

the deep learning model CNN were employed. Performance comparisons were conducted on the calibration, validation, and test sets. The detailed results are shown in Table 2 and Table 3, where the best-performing results for each task are highlighted in bold.

Table 2 shows that, for the classification task, the traditional machine learning models achieved better performance than the CNN model. Among them, SVC achieved accuracies of 0.81, 0.77, and 0.75 on the calibration, validation, and test sets, respectively, showing the best overall classification performance. Notably, the accuracies of all classification models were above 0.68 across the three datasets, indicating that the spectral preprocessing and modeling scheme provided effective discrimination of wheat varieties. For the regression tasks, the results in Table 3 show that PLSR achieved the highest  $r$  values for both SPAD and

LMC prediction among the single-task regression models. For SPAD prediction, the  $r$  values of PLSR on the calibration, validation, and test sets were 0.86, 0.80, and 0.83, respectively. For LMC prediction, the corresponding  $r$  values were 0.87, 0.81, and 0.84, respectively. The  $r$  values of the other regression models were also above 0.74, indicating relatively stable predictive performance across different datasets.

### 3.3.2. Results of the multi-task model

In this study, three types of multi-task models, including MLT-SVM, MLT-PLS, and MLT-CNN, were constructed to jointly model wheat variety classification, SPAD prediction, and LMC prediction. For MLT-CNN, three loss-weighting strategies, namely EWS, ARWS, and UWS, were further compared to evaluate their effects on task balance and

**Table 4**  
Results of the multi-task models.

| Model          | Dataset         | Classification | Regression  |      |             |      |
|----------------|-----------------|----------------|-------------|------|-------------|------|
|                |                 | accuracy       | $r_1$       | RMSE | $r_2$       | RMSE |
| MLT-SVM        | Calibration Set | <b>0.78</b>    | 0.84        | 2.59 | 0.85        | 1.00 |
|                | Validation Set  | <b>0.78</b>    | 0.79        | 2.78 | 0.82        | 0.99 |
|                | Test Set        | <b>0.75</b>    | 0.82        | 2.87 | 0.80        | 1.30 |
| MLT-PLS        | Calibration Set | 0.76           | 0.86        | 2.43 | 0.85        | 1.02 |
|                | Validation Set  | 0.75           | <b>0.81</b> | 2.62 | 0.82        | 1.01 |
|                | Test Set        | <b>0.75</b>    | 0.82        | 2.83 | <b>0.83</b> | 1.19 |
| MLT-CNN + EWS  | Calibration Set | 0.76           | <b>0.89</b> | 2.23 | <b>0.87</b> | 0.98 |
|                | Validation Set  | 0.73           | 0.79        | 2.81 | <b>0.85</b> | 1.00 |
|                | Test Set        | 0.69           | <b>0.84</b> | 2.74 | 0.81        | 1.30 |
| MLT-CNN + ARWS | Calibration Set | 0.73           | 0.86        | 2.68 | 0.84        | 1.58 |
|                | Validation Set  | 0.71           | 0.80        | 3.03 | 0.84        | 1.45 |
|                | Test Set        | 0.68           | <b>0.84</b> | 2.74 | 0.82        | 1.65 |
| MLT-CNN + UWS  | Calibration Set | <b>0.78</b>    | 0.88        | 2.54 | <b>0.87</b> | 1.05 |
|                | Validation Set  | 0.70           | 0.80        | 3.00 | 0.83        | 1.09 |
|                | Test Set        | 0.70           | 0.82        | 2.85 | 0.81        | 1.34 |

Note:  $r_1$  is the  $r$  of SPAD,  $r_2$  is the  $r$  of LMC.

model performance. The performance of each model on the calibration, validation, and test sets is shown in Table 4, where the best-performing results among different models are highlighted in bold.

Overall, the three multi-task models achieved good performance on the calibration set. MLT-SVM and MLT-PLS showed relatively similar performance in both classification and regression tasks. Among the traditional machine learning-based multi-task models, MLT-PLS achieved relatively good overall performance, with  $r$  values of 0.82 and 0.83 for SPAD and LMC prediction, respectively, and a classification accuracy of 0.75 on the test set.

For the MLT-CNN models, the EWS strategy achieved its optimal performance at the 1944th epoch, with an LR of 0.00008590. This model showed a favorable balance among the three tasks on the test set, with a classification accuracy of 0.69 and  $r$  values of 0.84 and 0.81 for SPAD and LMC prediction, respectively. These results indicate that the MLT-CNN with EWS provided the best overall task balance among the MLT-CNN models. In comparison, the UWS strategy achieved its best performance at the 1974th epoch, with an LR of 0.00006872,  $\sigma_{cls}$  of 1.0818,  $\sigma_{reg1}$  of 2.4377, and  $\sigma_{reg2}$  of 0.9953. This strategy slightly improved the test set classification accuracy to 0.70, but the regression performance decreased slightly, with  $r_1 = 0.82$  and  $r_2 = 0.81$ , suggesting that the optimization process may have been relatively biased toward the classification task. Under the ARWS strategy, the optimal performance was achieved at the 1399th epoch, with an LR of 0.00020972,  $\alpha$  of 0.2512,  $\beta$  of 0.3159, and  $\gamma$  of 0.4329. The MLT-CNN with ARWS exhibited comparable regression performance to the EWS and UWS strategies, but its classification accuracy was slightly lower at 0.68 on the test set.

In terms of loss values, the MLT-CNN with UWS obtained the lowest losses on the validation and test sets, with values of 2.90 and 3.22, respectively. The MLT-CNN with EWS achieved the lowest loss on the calibration set, with a value of 2.38. The loss values of the EWS and UWS strategies across the three datasets showed relatively small differences, suggesting stable performance across the calibration, validation, and test sets. In contrast, the loss values of the ARWS strategy were higher than those of the other two strategies, reaching 12.05, 12.74, and 12.53 on the calibration, validation, and test sets, respectively. Therefore, under the current task setting, ARWS was less suitable for direct application in loss weighting for the MLT-CNN model. From the perspective of performance distribution across the three tasks, ARWS showed weaker overall task balance than EWS and UWS.

Across all multi-task models evaluated, the MLT-CNN with EWS showed the most balanced overall performance across classification and regression tasks. It achieved the highest number of best-performing evaluation metrics among the multi-task models, indicating that shared convolutional feature extraction combined with equal task weighting was effective for coordinating multiple task objectives in this study.

### 3.3.3. Comparison of results between single-task models and multi-task models

For the classification task, among the single-task models, SVC achieved the best overall performance, with accuracies of 0.81, 0.77, and 0.75 on the calibration, validation, and test sets, respectively. In comparison, the MLT-CNN with EWS achieved accuracies of 0.69, 0.73, and 0.69 on the calibration, validation, and test sets, respectively. Although the classification performance of the multi-task model was lower than that of the single-task SVC model, it still maintained a comparable level of performance while simultaneously performing SPAD and LMC prediction. This result indicates that the multi-task model retained a certain degree of classification capability while integrating regression tasks within the same framework.

For the regression tasks, the single-task PLSR model showed good performance. For SPAD and LMC, it achieved  $r$  values of 0.86 and 0.87 on the calibration set, 0.80 and 0.81 on the validation set, and 0.83 and 0.84 on the test set, respectively. In comparison, the MLT-CNN with EWS yielded  $r$  values of 0.89 and 0.87 on the calibration set, 0.79 and 0.85 on

the validation set, and 0.84 and 0.81 on the test set for SPAD and LMC, respectively. For SPAD prediction, the MLT-CNN with EWS achieved a slightly higher test-set  $r$  value than the single-task PLSR model. In contrast, for LMC prediction, its performance was slightly lower than that of PLSR on the test set. This difference may be related to the different modeling mechanisms of PLSR and MLT-CNN. PLSR constructs a global linear regression model in a low-dimensional latent space by projecting spectra onto components that maximize covariance with the target variable. By contrast, the MLT-CNN with EWS uses one-dimensional convolutional layers and nonlinear activation functions to extract local spectral patterns and higher-order nonlinear features. Its shared feature extractor is jointly optimized across classification and regression tasks, allowing common spectral representations to be learned across different objectives. These results suggest that the multi-task model can achieve competitive regression performance while jointly modeling multiple wheat leaf phenotypic traits.

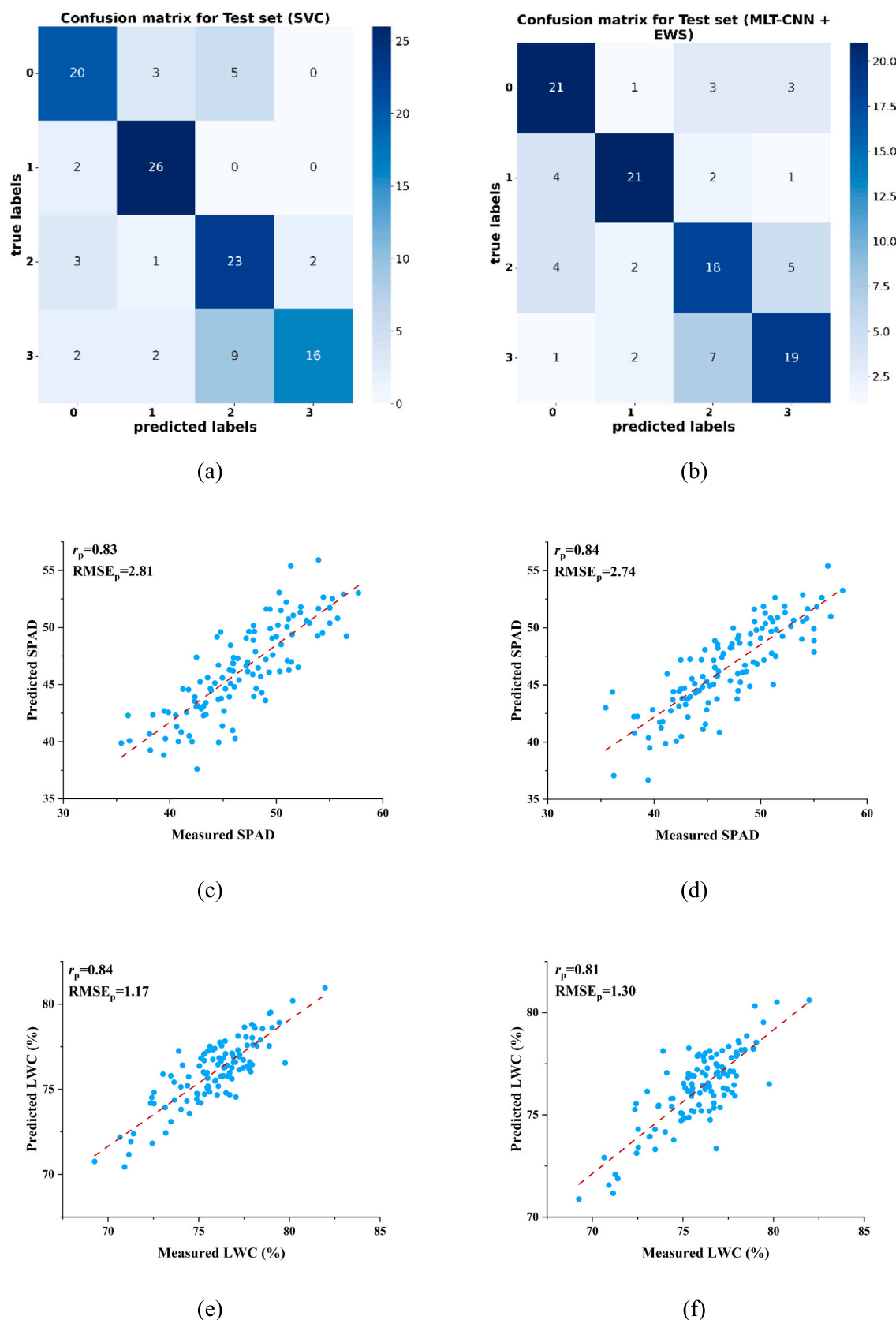
Overall, the MLT-CNN with EWS enabled shared feature learning and joint optimization across tasks while simultaneously handling classification and regression tasks. Although the MLT-CNN with EWS did not outperform the best single-task models in every metric, it achieved a slightly higher test-set  $r$  value for SPAD prediction and maintained relatively balanced performance across classification, SPAD prediction, and LMC prediction, while reducing the need to construct separate models for each task. Fig. 4 shows the confusion matrices and scatter plots of the test sets for the optimal single-task and multi-task models.

### 3.4. Wavelength contribution patterns revealed by SHAP

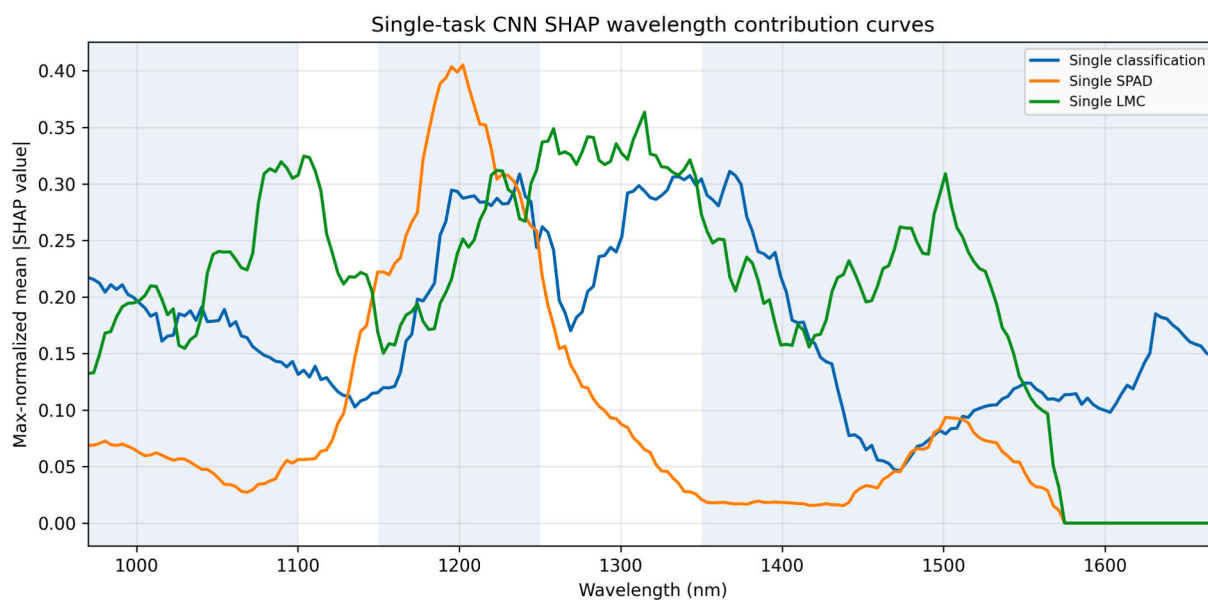
Although the above results demonstrate the predictive feasibility of single-task and multi-task models, performance metrics alone cannot explain which wavelength regions contributed most to each task or how the MLT-CNN used shared and task-specific spectral information. Therefore, to further interpret the spectroscopic basis of the CNN-based models, SHAP-based wavelength contribution analysis was conducted for the single-task CNNs and the three branches of the MLT-CNN. Similar SHAP-based wavelength interpretation has been used in recent hyperspectral deep-learning studies to identify representative and physiologically meaningful spectral regions [28]. Fig. 5(a) shows the SHAP wavelength contribution curves of the single-task CNNs for wheat variety classification, SPAD prediction, and LMC prediction, while Fig. 5(b) shows the branch-specific SHAP contribution curves of the MLT-CNN.

For the single-task CNNs, the three tasks exhibited different wavelength contribution patterns. The classification model showed relatively broad contributions across several wavelength regions, including approximately 970–1050 nm, 1190–1240 nm, 1280–1380 nm, and 1630–1660 nm. This result indicates that wheat variety classification relied on multiple spectral regions, which may be associated with cultivar-dependent differences in leaf structure, scattering characteristics, water status, pigment-related physiological variation, and biochemical composition. The SPAD prediction model showed a concentrated high-contribution region around 1150–1240 nm. This region is related to O—H and C—H overtone or combination absorption and may indirectly reflect leaf structural, water-related, and biochemical variations associated with chlorophyll-related SPAD changes [29,30]. For LMC prediction, relatively high contributions were observed mainly in the short- to middle-SWIR region, including approximately 1050–1120 nm, 1200–1350 nm, and 1450–1530 nm. These regions suggest that LMC prediction was influenced not only by water-sensitive O—H absorption features, but also by leaf structure and continuous spectral-shape information related to water status [31].

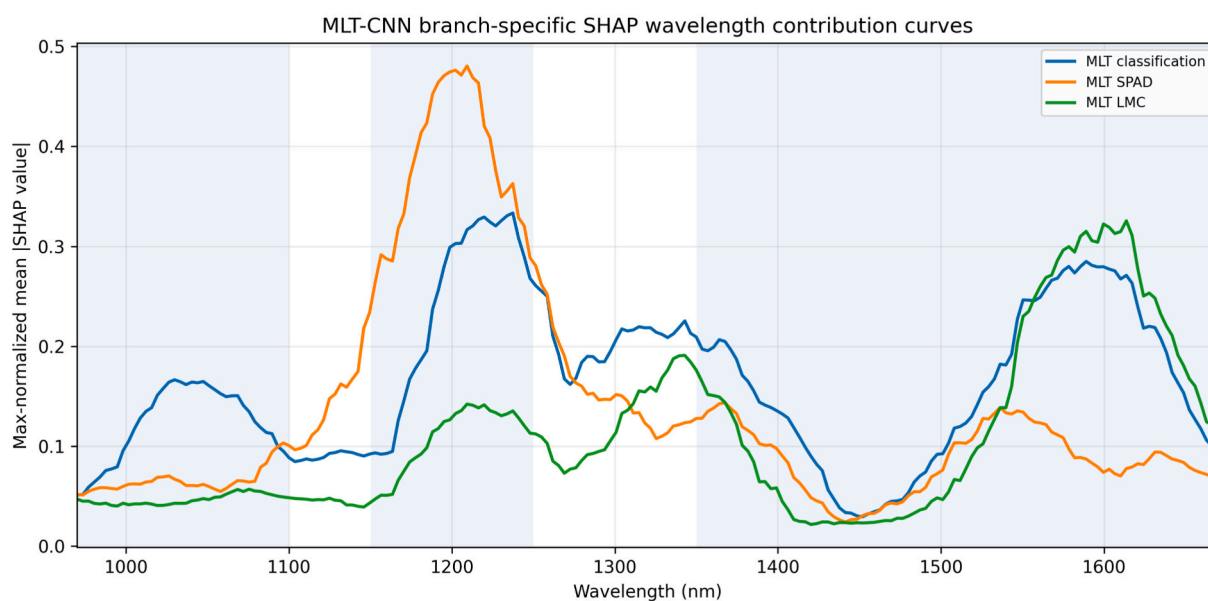
It should be noted that the single-task regression CNNs showed near-zero SHAP values in the terminal long-wavelength region. This phenomenon was mainly related to the layer-wise downsampling behavior of the regression CNN architecture. Specifically, the repeated convolutional operations progressively compressed the spectral dimension and



**Fig. 4.** Confusion matrices and scatter plots of the test sets for the optimal single-task and multi-task models. (a) Confusion matrix for wheat variety classification using single-task SVC; (b) confusion matrix for wheat variety classification using MLT-CNN + EWS; (c) scatter plot for SPAD prediction using single-task PLSR; (d) scatter plot for SPAD prediction using MLT-CNN + EWS; (e) scatter plot for LMC prediction using single-task PLSR; (f) scatter plot for LMC prediction using MLT-CNN + EWS.



(a)



(b)

**Fig. 5.** SHAP-based wavelength contribution analysis of the CNN-based models. (a) Wavelength contribution curves of the single-task CNNs for wheat variety classification, SPAD prediction, and LMC prediction. (b) Branch-specific wavelength contribution curves of the MLT-CNN. The curves represent max-normalized mean absolute SHAP values across the test samples. The shaded regions indicate typical SWIR spectral intervals related to leaf structural scattering, O—H/C—H-associated absorption, water-sensitive O—H absorption, and O—H/N—H/C—H combination absorption.

weakened the effective attribution pathway from the terminal input wavelengths to the final regression outputs. Therefore, the near-zero SHAP values in this region should be interpreted as model-specific attribution results of the single-task regression CNNs, rather than as evidence that the terminal long-SWIR region lacks water- or biochemical-related spectral information. This distinction is important because the MLT-CNN still assigned evident contributions to the 1550–1640 nm region, particularly in the LMC and classification branches, indicating that this long-SWIR region remained informative when modeled within the shared multi-task framework [32,33].

For the MLT-CNN, the three branches showed both shared and task-specific wavelength contribution patterns. The SPAD branch had its strongest response around 1150–1240 nm, which was consistent with the single-task SPAD model and indicates that this wavelength region provided stable SPAD-related spectral information. The classification branch also showed evident contributions in the 1180–1240 nm region, suggesting that this region contained shared spectral information useful for both cultivar discrimination and SPAD-related estimation. In contrast, the LMC branch exhibited stronger contributions in the 1550–1640 nm region. This region is associated with O—H, N—H, and

C—H combination absorption and may reflect water status, protein, starch, cellulose, and other biochemical constituents [34]. The classification branch also showed increased contribution in this long-SWIR region, indicating that cultivar discrimination may also benefit from water- and biochemical-related spectral differences.

Overall, the SHAP results indicate that the CNN-based models did not rely on a single wavelength interval. Instead, different tasks used both shared and task-specific spectral information. The 1150–1240 nm region contributed strongly to SPAD prediction and also supported classification, suggesting a shared spectral basis related to structural, water-related, and biochemical responses. The 1550–1640 nm region was more prominent in the MLT-CNN, especially for the LMC and classification branches, indicating that the multi-task model enhanced the use of long-SWIR information related to water status and biochemical constituents. These results provide a spectroscopic explanation for the MLT-CNN structure, in which the shared convolutional feature extraction module captures common spectral representations, while the task-specific branches retain task-dependent wavelength responses.

#### 4. Discussion

The significance of this study lies in exploring the feasibility of integrating hyperspectral imaging with multi-task learning for the non-destructive assessment of wheat leaf phenotypic traits. SPAD and LMC are key physiological indicators related to chlorophyll status, photosynthetic potential, water status, and stress response, all of which are closely associated with crop growth and yield formation. Rapid and accurate acquisition of these indicators can provide important support for crop health diagnosis and precision management. Therefore, integrating SPAD prediction, LMC prediction, and wheat variety classification within the same spectral modeling framework is not only scientifically valuable but also practically significant for wheat growth monitoring and precision management. In addition, the use of wavelength-level interpretation in this study further helped connect model predictions with the spectroscopic information contained in wheat leaf spectra.

Classification of wheat varieties based on leaf hyperspectral characteristics was found to be feasible. Although the overall spectral profiles of different varieties were generally similar, cultivar-related differences in reflectance magnitude and absorption features at specific wavelengths still provided useful information for variety discrimination. Previous studies have explored leaf-based crop classification using hyperspectral imaging, such as classifying young leaves of *Toona sinensis* with UAV-based hyperspectral imagery [35], identifying rice varieties based on leaf hyperspectral features [36], and classifying tree species across different seasons and regions using leaf hyperspectral images [37]. In this study, experiments were conducted on four wheat varieties, and the single-task SVC model achieved an accuracy of 0.75 on the test set, indicating effective discrimination of wheat varieties based on leaf spectral information. The SHAP results further suggested that variety classification relied on multiple spectral regions rather than a single wavelength interval, which is consistent with the fact that cultivar differences may arise from combined variation in leaf structure, water regulation, pigment-related physiological status, and biochemical composition.

Regarding SPAD and LMC prediction, the results of this study are generally consistent with previous hyperspectral studies on wheat physiological trait estimation. In this study, the single-task PLSR model achieved a coefficient of determination ( $R^2$ ) of 0.69 for SPAD on the test set. This result is comparable to the  $R^2$  value of 0.61 reported for a random forest regression (RFR) model in a previous study on rapid and non-destructive wheat SPAD assessment using hyperspectral imaging [30]. For LMC prediction, the PLSR model in this study achieved an  $R^2$  value of 0.71 on the test set. Similarly, a previous study using

hyperspectral technology to predict nitrogen and water content in wheat reported an  $R^2$  value of 0.67 for wheat moisture content using PLSR [38]. These comparable results further support the scientific validity of the method adopted in this study. From the perspective of spectral interpretation, SPAD-related contributions were mainly concentrated around 1150–1240 nm. For LMC, the contribution patterns involved water-sensitive and biochemical-related SWIR regions, indicating that LMC prediction was influenced not only by water absorption but also by tissue structure and continuous spectral-shape information.

Existing studies have explored the integration of hyperspectral imaging and multi-task learning in different spectral analysis tasks. For example, collaborative representation combined with multi-task learning has been used for hyperspectral image classification [39], and multi-task learning has also been applied to spatial hyperspectral unmixing [40]. In plant phenotyping and agricultural spectral analysis, related studies have included the simultaneous prediction of SPAD and water content in rice leaves using hyperspectral imaging, deep multi-task regression, and transfer component analysis [14], as well as wolfberry origin identification using hyperspectral imaging and multi-task residual fully convolutional networks [41]. Multi-task deep learning has also been extended to near-infrared spectral analysis, such as food origin identification and component prediction based on S-transform and multi-task deep learning [42], and multi-task analysis of Hongmeiren citrus using near-infrared spectra and machine learning [43]. These studies indicate that multi-task learning has potential for integrating multiple spectral-analysis objectives. However, most existing studies focus on either classification tasks, multiple regression tasks, or origin classification combined with component prediction, whereas simultaneous wheat variety classification, SPAD prediction, and LMC prediction from leaf hyperspectral spectra remains less explored. To address this gap, this study integrated wheat variety classification, SPAD prediction, and LMC prediction into a unified leaf-level hyperspectral multi-task modeling framework. In addition to evaluating predictive performance, SHAP-based wavelength contribution analysis was further introduced to interpret the shared and task-specific spectral information used by the CNN-based models, thereby extending the analysis from model comparison to wavelength-level spectroscopic interpretation.

Notably, classification and regression tasks differ substantially in optimization objectives, convergence behavior, loss scale, and gradient backpropagation. For example, Jiang et al. [43] used a feedforward neural network combined with Savitzky-Golay filtering and Boruta feature selection to perform origin classification and chemical-element analysis of Hongmeiren citrus, showing the feasibility of combining classification and quantitative prediction in spectral analysis. In this study, the MLT-CNN adopted a shared one-dimensional convolutional encoder and task-specific branches to analyze wheat leaf hyperspectral data. This structure allowed the model to learn shared spectral representations while maintaining separate outputs for wheat variety classification, SPAD prediction, and LMC prediction. The results showed that the MLT-CNN with EWS achieved relatively balanced performance across the three tasks, supporting the suitability of this model design for integrated wheat leaf phenotyping.

The SHAP results provided further support for this model design from a spectroscopic perspective. This is consistent with recent explainable multi-task learning studies, in which SHAP was used to dissect trained multi-task models and identify task- or group-related feature relevance [44]. The 1150–1240 nm region was strongly related to SPAD prediction and also contributed to variety classification, suggesting that this region contained shared spectral cues associated with O—H/C—H-related absorption, leaf structure, water-containing components, and biochemical variation [45]. In contrast, the MLT-CNN assigned stronger contributions to the 1550–1640 nm region in the LMC and classification branches, indicating that long-SWIR information related to O—H, N—H, and C—H combination absorption may

contribute to water status and cultivar-dependent biochemical differences [46]. These results suggest that the MLT-CNN did not simply combine three output tasks within one network, but captured both shared and task-specific wavelength information. This provides a spectroscopic explanation for using a shared convolutional encoder together with task-specific branches.

Loss function design is critical in multi-task learning because it determines how different task losses are balanced during optimization. Even a well-designed model architecture may underperform if the task losses are not appropriately weighted. Therefore, this study compared three loss-weighting strategies, including EWS, ARWS, and UWS proposed by Kendall et al. [24]. The UWS strategy achieved the lowest total loss on the validation and test sets, suggesting its advantage in optimizing the combined loss objective. However, this did not necessarily lead to the best overall task balance. The MLT-CNN with EWS achieved more balanced performance across classification and regression tasks, particularly by maintaining competitive regression performance while retaining acceptable classification accuracy. This phenomenon may be related to the sample size and task heterogeneity in this study. UWS relies on the dynamic estimation of task uncertainty, and such estimation may be unstable when the dataset is relatively limited. In contrast, EWS avoids over-reliance on adaptive weight estimation and showed better robustness under the current data conditions. This observation is consistent with Shi et al. [42], who also adopted an equal-weighting strategy in multi-task deep learning for food origin identification and component prediction, supporting the feasibility of equal weighting in spectral multi-task modeling. For ARWS, the loss scales of classification and regression tasks differed substantially; therefore, linear weighting with  $\alpha$ ,  $\beta$ , and  $\gamma$  may have caused one task weight to dominate, making the reciprocal penalty terms unstable and leading to weaker overall task balance.

This study compared the performance of single-task and multi-task models and showed that the multi-task model did not outperform the best single-task models in every task. Specifically, for the classification task, the single-task SVC model achieved a test-set accuracy of 0.75, which was higher than that of the MLT-CNN with EWS (0.69). For the regression tasks, the performance varied between SPAD and LMC prediction. The single-task PLSR model achieved a test-set  $r$  value of 0.83 for SPAD prediction, whereas the MLT-CNN with EWS achieved a slightly higher  $r$  value of 0.84. In contrast, for LMC prediction, PLSR achieved a test-set  $r$  value of 0.84, which was higher than that of the MLT-CNN with EWS (0.81). These results indicate that the multi-task model achieved performance comparable to that of the best single-task models in regression tasks, but did not consistently surpass them across all targets. This may be attributed to task interference in shared representation learning, because wheat variety classification, SPAD prediction, and LMC prediction may rely on different spectral features, optimization objectives, and gradient-update directions. The SHAP results support this explanation by showing that different tasks had partially different wavelength dependencies. Therefore, although shared spectral representations can improve learning efficiency and benefit some tasks, they may also limit performance improvements in other tasks under certain conditions [4]. This explains why the MLT-CNN achieved balanced multi-task performance but did not consistently exceed the best single-task models for all individual targets.

Overall, the contribution of this study is not limited to constructing predictive models for wheat leaf phenotyping. By combining hyperspectral imaging, multi-task learning, and SHAP-based wavelength contribution analysis, this study showed that wheat leaf spectra contain both shared and task-specific information for cultivar classification, SPAD prediction, and LMC prediction. The shared information supports the use of a common spectral feature extractor, whereas the task-specific wavelength responses justify the use of separate output branches. Therefore, the proposed framework provides not only an integrated

modeling strategy, but also a more interpretable spectroscopic perspective for multi-trait wheat leaf phenotyping.

This study also has several limitations. First, the experimental dataset was limited to four wheat varieties, and the sample size should be further expanded in future studies. Different wheat leaves have inherent physical and chemical characteristics that vary, such as tissue structure, light absorption rate and chemical composition. Therefore, the robustness and broader applicability of the proposed models still require further validation using larger and more diverse datasets. In addition, this study focused on developing and comparing single-task and multi-task modeling strategies for wheat variety classification, SPAD prediction, and LMC prediction using one batch of experimental data. In future work, samples collected from multiple growth stages, different seasons, and independent experimental batches should be included to evaluate the stability and transferability of the models. Second, classification and regression tasks differ in task difficulty, optimization objectives, loss scales, and feature requirements, which may lead to task imbalance or task interference in multi-task learning. Therefore, in practical applications, the suitability of multi-task learning should be evaluated according to the specific target traits, data scale, and application scenario. Third, although SHAP analysis provided wavelength-level interpretation of the CNN-based models, the identified contribution patterns were model-specific and should be further validated using larger datasets, independent experimental batches, and complementary spectroscopic interpretation methods. Future studies may also combine SHAP with characteristic wavelength selection, variable importance analysis, and biochemical reference measurements to further clarify the physiological basis of the important wavelength regions identified in this study.

## 5. Conclusions

This study developed a multi-task learning framework based on hyperspectral imaging to integrate wheat variety classification, SPAD prediction, and LMC prediction within a unified spectral modeling process. Both single-task and multi-task models were constructed and compared. In the single-task models, SVC achieved the best classification performance, with an accuracy of 0.75 on the test set, while PLSR performed best in the regression tasks, with  $r$  values of 0.83 for SPAD and 0.84 for LMC. In the multi-task models, the MLT-CNN with EWS achieved balanced performance across the three tasks, with a test-set classification accuracy of 0.69, an  $r$  value of 0.84 for SPAD, and an  $r$  value of 0.81 for LMC. Although the MLT-CNN with EWS did not outperform the best single-task models in every metric, it achieved a slightly higher test-set  $r$  value for SPAD prediction and provided an integrated framework for simultaneous classification and regression.

SHAP analysis further indicated that the CNN-based models captured both shared and task-specific wavelength contribution patterns related to cultivar identity, chlorophyll-related SPAD variation, and water-related LMC variation, thereby providing wavelength-level spectroscopic evidence for the interpretability of the proposed multi-task hyperspectral framework. The shared spectral information supports the use of a common one-dimensional convolutional encoder, whereas the task-specific wavelength responses justify the use of separate task branches in the MLT-CNN. Overall, hyperspectral imaging combined with multi-task learning provides a feasible and interpretable strategy for multi-trait crop monitoring. Future research should include larger datasets covering more wheat varieties, multiple growth stages, different seasons, and independent experimental batches. In addition, complementary spectral interpretation methods, such as characteristic wavelength selection, variable importance analysis, attention mechanisms, and biochemical reference measurements, could be incorporated to further validate the physiological basis of the key wavelength regions identified in this study.

## CRediT author statement

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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